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192312016

```
1)
P_A = 1/4
P_B = 1/2
P_C = 1/4

P_Blue_A = 4 / (6 + 4 + 2)
P_Blue_B = 7 / (5 + 7 + 3)
P_Blue_C = 2 / (8 + 2 + 5)

P_Blue = (
    P_Blue_A * P_A +
    P_Blue_B * P_B +
    P_Blue_C * P_C
)

P_C_given_Blue = (P_Blue_C * P_C) / P_Blue

print("Probability that the selected box was Box C:")
print(P_C_given_Blue)

print("\nRounded to 4 decimal places:")
print(round(P_C_given_Blue, 4))
```

output

Probability that the selected box was Box C:
0.09523809523809525

Rounded to 4 decimal places:
0.0952

```
2)

P_A = 1/4
P_B = 1/2
P_C = 1/4

P_Blue_A = 3 / (5 + 3 + 2)
P_Blue_B = 6 / (4 + 6 + 5)
P_Blue_C = 2 / (7 + 2 + 6)

P_Blue = (
    P_Blue_A * P_A +
    P_Blue_B * P_B +
    P_Blue_C * P_C
)
```

```
P_C_given_Blue = (P_Blue_C * P_C) / P_Blue
```

```
print("Probability that the car came from Section C =", P_C_given_Blue)
print("Rounded to 4 decimal places =", round(P_C_given_Blue, 4))
```

output

```
Probability that the car came from Section C = 0.1081081081081081
Rounded to 4 decimal places = 0.1081
```

3)

```
TP = 210
TN = 190
FP = 30
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```
total_samples = 1800 * 2
test_samples = total_samples * 0.20
```

```
FN = int(test_samples - (TP + TN + FP))
```

```
accuracy = (TP + TN) / test_samples
precision = TP / (TP + FP)
sensitivity = TP / (TP + FN)
specificity = TN / (TN + FP)
```

```
print("False Negatives (FN) =", FN)
print("Accuracy =", round(accuracy, 4))
print("Precision =", round(precision, 4))
print("Sensitivity (Recall) =", round(sensitivity, 4))
print("Specificity =", round(specificity, 4))
```

output;

```
False Negatives (FN) = 290
Accuracy = 0.5556
Precision = 0.875
Sensitivity (Recall) = 0.42
Specificity = 0.8636
```

4)

```
P_X = 1/3
P_Y = 2/3
```

```
P_Adulterated_X = 30 / (20 + 30)
P_Adulterated_Y = 20 / (40 + 20)
```

```

P_Adulterated = (
    P_Adulterated_X * P_X +
    P_Adulterated_Y * P_Y
)

P_X_given_Adulterated = (P_Adulterated_X * P_X) / P_Adulterated

print("Probability that the bottle came from Shop X =",
      P_X_given_Adulterated)
print("Rounded to 4 decimal places =", round(P_X_given_Adulterated, 4))

```

output

Probability that the bottle came from Shop X = 0.4736842105263158
 Rounded to 4 decimal places = 0.4737

5)

```

P_A = 3/4
P_B = 1/4

```

```

P_Defective_A = 15 / (15 + 35)
P_Defective_B = 25 / (25 + 25)

```

```

P_Defective = (
    P_Defective_A * P_A +
    P_Defective_B * P_B
)

```

```

P_A_given_Defective = (P_Defective_A * P_A) / P_Defective

```

```

print("Probability that the defective item came from Section A =",
      P_A_given_Defective)
print("Rounded to 4 decimal places =", round(P_A_given_Defective, 4))

```

output

Probability that the defective item came from Section A =
 0.6428571428571428
 Rounded to 4 decimal places = 0.6429